

Limitations of Analyzing Anonymous, Self-Reported Data

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Every number in this analysis describes the 40,000 Grad Café entries in my database, not graduate applicants in general, and the gap between those two things is where the limitations lie. Grad Café is anonymous and voluntary, so it suffers from selection bias: only people who know the site exists, are comfortable posting, and choose to report a result appear in it, and the people who are absent (applicants who never visit the site, who come from fields or countries where posting results is uncommon, or who simply prefer privacy) are not a random slice of the population. The clearest example in my own results is Question 3: the average GRE Quantitative score is 165.68, but only 1,176 of the 40,000 entries (2.94%) report one. A commonly cited average for all GRE test takers is roughly 157, so it would be a mistake to conclude that applicants in general average 165.68. People with strong scores are more likely to volunteer them, many programs no longer require the GRE so weaker scorers have little reason to post theirs, and the site's audience may skew toward highly competitive applicants. The missing values are therefore not missing at random, and averaging only the people who report a metric (as the assignment requires) describes the reporters, not the applicant pool. Self-reported values are also unverified and messy: in the raw data the great majority of GRE writing scores were a 0.0 placeholder for "not provided", some GPAs were below 1.0, and 847 entries had a meaningless nationality value, all of which I had to convert to missing values before any average was meaningful.

Outcome reporting adds a second layer of bias. People with especially good news (an acceptance from a top program) or especially bad or surprising news (a rejection they consider unfair, a strange timeline) are more motivated to post than people with an unremarkable result, so the dataset over-represents memorable outcomes. My Fall 2025 acceptance percentage in Question 5 is 39.03%; that is the share of Grad Café entries that were acceptances, not the acceptance rate of the programs involved, which are often published as much lower for selective programs. Another result that needs care is my second original question: among Fall 2026 applicants the accepted group has an average GPA of 3.76 and GRE Quantitative score of 165.61, compared with 3.78 and 165.59 for the rejected group. The groups look almost identical, but that does not show that grades and test scores are irrelevant to admission: everyone who reports scores here is already clustered in a narrow, high range, decisions depend on things the database never captures (letters, research, fit, funding), and reporting habits may differ by outcome. The pipeline adds error as well: LLM standardization exists for 10,442 entries and makes occasional mistakes, so the Question 8 and Question 9 counts should be treated as approximate. What my database mathematically contains is a set of self-selected, partly missing, unverified reports; the

defensible conclusions are descriptive ("among people who chose to post, ...") rather than statements that are representative of all graduate applicants.